



voestalpine VAE VKN India Pvt. Ltd.

PROJECT: EDFCC APL-01



STORAGE, HANDLING, ASSEMBLING, INSTALLATION & MAINTENANCE MANUAL
FOR
1IN12-555R-60E1 CANTED TANGENTIAL TURNOUT WITH 60E1A1 THICK WEB
SWITCHES B.G. (1676MM) ON PSC SLEEPERS FOR 25T AXLE LOAD
(LOT- 101, 102 & 103).

Version 01	Name	Department	Date	Signature
Created	Sandeep Paul	P&C	19.12.2017	
Checked	Biraj Deb	Technical Bureau	19.12.2017	
Approved	Biraj Deb	Technical Bureau	19.12.2017	

VERSION HISTORY:

Version No.	Modified	Date	Approved	Comment
03	As per DFCC requirement	12.05.2020		

COPIES DISTRIBUTED TO:

Version No.	Company	Copies	Date	Comment

Reproduction, distribution and utilization of this document as well as the communication of its contents without voestalpine VAE GmbH's explicit authorization are prohibited. Offenders will be held liable for the payment of damages. All rights reserved in the event of the grant of a patent, utility model or design. Copyright reserved.

CONTENT	PAGE
PART - A	
1. INTRODUCTION	4
2. SALIENT FEATURES	5-6
3. STORAGE OF TURNOUT ITEMS	7-8
4. HANDLING OF TURNOUT ITEMS	9-10
5. ASSEMBLY OF TURNOUT ITEMS	11-30
6. INSTALLATION OF TURNOUT ITEMS	31-35
6.1. Step-by-step: Installation Procedure for 1in12 T/O	36-37
7. DESCRIPTION OF TURNOUT	38-39
PART B	
8. GENERAL RECOMMENDATION AND USE OF THIS HANDBOOK	41
9. FAILURE DEFINITION	42
9.1. Rail crack/ cross break	42
9.2. Large surface defects	43
9.3. Cracks in the rail heads	43
9.4. RCF of running surface	43
9.5. Critical breakouts on the running edge	44
9.6. Small running edge defects	44
9.7. Pollution of sliding chairs	44
9.8. Heavy deformed plastic pads/ break of plastic pads	44
9.9. Broken base/ slide plates	45
10. WEAR LIMITS	46-55
11. PREVENTIVE MAINTENANCE	56-64
12. CORRECTIVE MAINTENANCE	65-70

1. INTRODUCTION

To keep turnouts in service, all parts forming the turnout have to be stored and handled accordingly as well as inspected periodically. These inspection intervals are the basics to guarantee a certain life span as well as a safe condition of the turnouts.

Point machines are not included in this manual as these aren't part of the delivery scope of VAEVKN.

This manual could only be used as addition to the internal regulations of EDFCC in regard to storage, handling, installation and maintenance and repair actions. There will be no references to EDFCC internal regulations.

Maintenance and inspection intervals depend mainly on the loading conditions of a turnout (train speeds, frequency, type of sub ground, etc.). The following intervals are just average values and have to be adapted if needed by the responsible authorities.

The manual is structured as follows:

Chapter 2: Describes the salient features of Canted Turnout

Chapter 3: Describes the methodology for storage of the turnout items

Chapter 4: Describes the ways of handling the turnout items

Chapter 5: General information regarding assembly of the turnouts

Chapter 6: Describes the installation

Chapter 7: Turnout description

Chapter 8: General information and

Chapter 9: Failure descriptions

Chapter 10: Wear Limits

Chapter 11: provides recommendations on how preventive maintenance has to be done to detect possible failures at a very early stage, and how to maintain them to keep the turnout for a longer life span in proper shape (service). This part of the documentation includes a list of failures as well as the actions that have to be done if one or more of the mentioned failures occur. In this chapter the turnout is split in the three main units – switch device, intermediate area and crossing area to allow a systematic inspection

Chapter 12: describes the corrective maintenance that has to be done when failures occur. Therefore simply find the component in this chapter and follow the instructions*

*Please note that the aforementioned instructions can only be seen as an additional recommendation to the internal regulations of the track operator.

2. SALIENT FEATURES

 VAE VKN		voestalpine VAE VKN India Pvt. Ltd.	
SALIENT FEATURES OF DESIGN FOR 1IN12-555R-60E1 CANTED TANGENTIAL TURNOUT WITH 60E1A1 THICK WEB SWITCHES B.G. (1676MM) ON PSC SLEEPERS FOR 25T AXLE LOAD TO DRAWING NO. VAEVKN04743-03			
SI No.	Salient Features		
A	Turnout		Comments
1	Switch (Drawing No. VAEVKN04756-03)	Axle Load	Suitable for 25 MT
2		Cant Deficiency	75 mm
3		Turnout Geometry	Tangential Entry
4		Design Speed for 1 in 12	55 Km/hr
5		Track Gauge	1676 mm
6		Rail inclination of the Turnout	1 in 20
7		Turnout Radius	555 METER
8		Tongue Rails	Turnout switches are made of Asymmetrical Thick-Web Tongue Rails (Zu-1-60 / 60E1A1) Grade: 1080HH / R350HT
9		Rail Machining	Machining done in CNC with surface Roughness less than 6 µm
10		Anti Creep	Special kind of Anti Cree Device installed to enable the turnout to put on CWR Track.
11		Back Drive	Torsion Type Back Drive is designed and installed with this turnout to achieve proper housing from A.T.S to the J.O.H of the switch and minimum J.O.H clearance of 60 ± 2.
12		Slide Chair	1. Slide Chairs are fabricated out of Mild Steel. 2. Slide chairs having elastic fastening. 3. Slide Chairs are provided with Pi- rolls on intermittent sleeper. 4. Sliding part of the Slide Chairs is coated with nickel-chromium plating.
13		Stock Rail	1. Stock Rail-- (60E1) shall be steel grade of 1080HH

14	Crossing (Drg no. VAEVKN05107-03)		1. All Crossing are weldable CMS Crossing manufactured from Austenitic Manganese Steel as defined in IRS T29-2016 2. All Rails for Weldable CMS Crossing are 1080 grade rail welded on it. 3. CMS Crossing Top Machining done in CNC machine having surface roughness less than 6 µm 4. Turnout Radius will continue through the CMS Crossing for LH & RH Turnouts 5. Rail inclination: 1 in 20
15	Check Rail		1. Check Rails made out of 33C1 with R260 grade. 2. Check rails shall be 20mm higher than rail table
16	Lead Area (Drg no. VAEVKN04743-03)	Rail inclination	1. Rail inclination: 1 in 20 2. Rail inclination provided with the help of canted Base Plates.
17	Sleepers		1. Turnout Sleepers are laid in Fan-shaped right from Sleeper no. 1 2. For RH & LH Turnouts, same Sleepers can be used.
18	Turnout	Fastening System	1. M.S. Canted and fabricated Base Plates are provided. 2. IRS Standard (Mark V as per RDSO Drg. No. RDSO/T-5919) fastenings. 3. 6 mm GRSP under Base Plates. 4. 10 mm Composite Rubber Pads between Rail and Base Plates. 5. Single coil spring washer RT-10773. 6. Plate Screw RT-3912 & RT-3913.
		Length of Turnout	48260 mm

3. STORAGE OF TURNOUT ITEMS



FIG-1: RAILS STACKING FOR EDFCC

Fastening Materials, pads, small equipment as well baseplates (if not mounted on switch device) are packed into transport boxes. Seaworthy packing for overseas delivery.

The turnout types as well as turnout numbers and colour code marking (if applicable) are visible on the labels of the wooden packing lists.



FIG-2 (a): PACKAGING of FASTENING ITEMS



FIG-2 (b): BASEPLATE STACKING FOR EDFCC



FIG-3: FITTINGS & FASTENING STACKING FOR EDFCC

3. HANDLING OF TURNOUT ITEMS

Rails shall be loaded in wagons in layers with wooden/steel spacer flats between them so that the rails are not damaged during transportation.
The rails shall be tied as per the extant instructions.

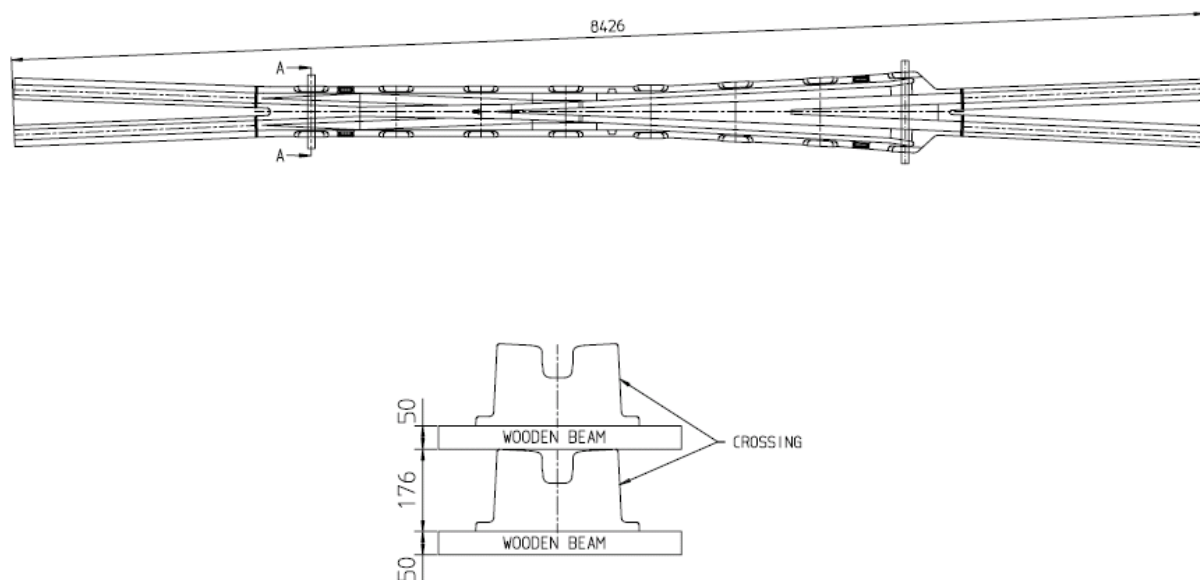


FIG-4: HANDLING OF CROSSINGS

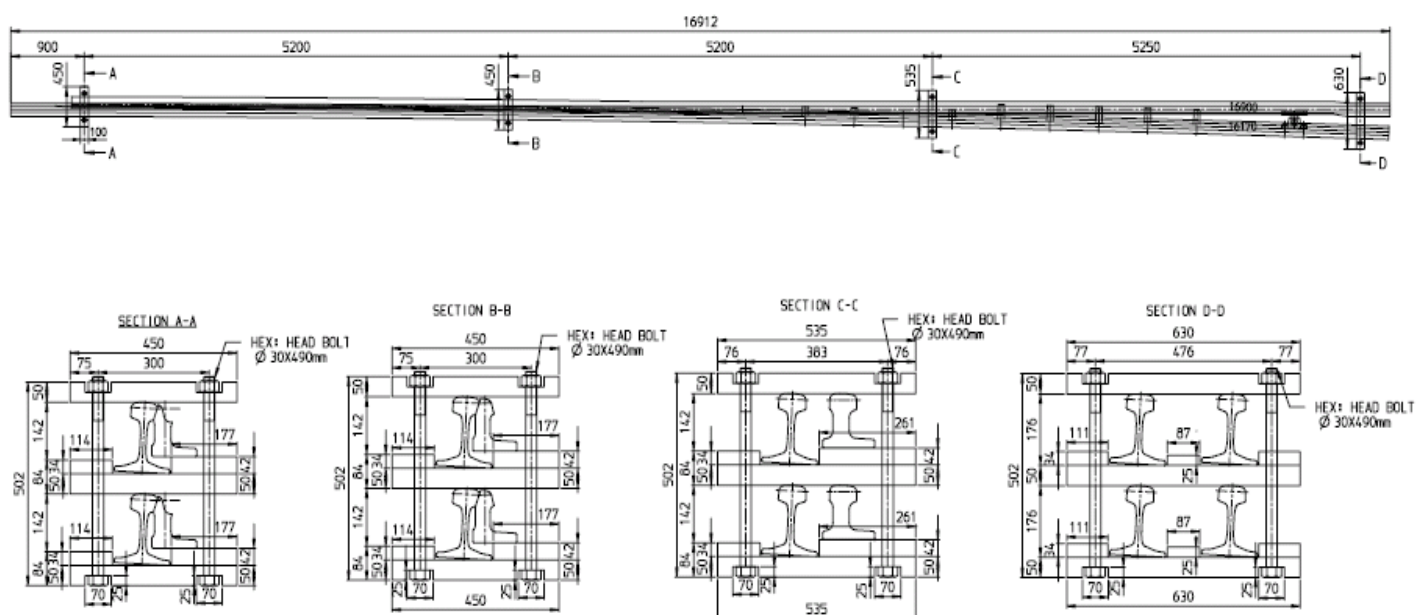


FIG-5: HANDLING OF SWITCH



FIG- 6 (a): TURNOUT HANDLING



FIG 6 (b): TURNOUT HANDLING

5. ASSEMBLY OF TURNOUT ITEMS

Sleeper No.	SLEEPER WISE FITTINGS IN SWITCHES FOR 1 IN 12 555 R CANTED TURNOUT						
	Description of fittings with Drawing No & Quantity						
	Standard B.P	Slide chairs	Machined Bearing plate	Combined Bearing plate	GRSP 10 mm	GRSP 06mm	Remarks
1	LH & RH: 05473: 02				04918:02	05483:02	Plate Screw RT- 3912/3913:0 4 each
2		LH & RH: 04928:0 & 07668:01			05493:02	05408:01 05409:02 07462:01	Plate Screw RT- 3912/3913:0 8 & 04, Tie Plate 05403:01 M.S. Plate 05405:01
3		LH & RH: 04923:02			04198:02	05484:02 05408:01 07462:01	Plate Screw RT- 3912/3913:0 8 & 04, M.S Plate 05405:01
4		LH & RH: 04923:02			04198:02	05484:02	i) Plate screw RT- 3912/3913:0 4 on sleeper no:4 to 14 & 17 to 28 each ii) All plate screws shall be provided with single coil spring washer(SCSW) T-10773 iii) Leaf spring 04195 & Wedge 04196:02 each on sleeper no:02 to 25 each iv) Piroll with spring loaded roller 05521: 02 on sleeper no:21 & 23
5		LH & RH: 04928:02			05493:02	05079:02	
6		LH & RH: 04923:02			04198:02	05484:02	
7		LH & RH: 04923:02			04198:02	05484:02	
8		LH: 04929:01 RH:04930:01			05493:02	05080:01 05491:01 07462:01	
9		LH: 04924:01 RH:04925:01			04198:02	05485:02	
10		LH: 04924:01 RH:04925:01			04198:02	05485:02	
11		LH: 04929:01 RH:04930:01			05493:02	05080:01 05491:01	
12		LH: 04924:01 RH:04925:01			04198:02	05485:02	
13		LH: 04929:01 RH:04930:01			05493:02	05080:01 05491:01 07462:01	

14		LH: 04924:01 RH:04925:01			04198:02	05485:02 07462:01	each v) Piroll with single roller 05520: 02 on sleeper no:17 & 19 each vi) Piroll with double roller 05519: 02 on sleeper no : 2,5,8,11,13,15 each vii) Liner RT-8222: 02 on sleeper no:01,26,27,28 each viii) Liner RT-8223:02 on sleeper no 1 to 28 ix) All ERC MK-V (RT-5919) shall be fixed along with liners RT-8222 and RT-8223 x) Back drive shall be fitted on opposite side of point M/C on SL no 3,8,13 & 14 on M.S plate 07453:04 and 07685:01 on sleeper no. 2. 04 plate screw RT-3912 for each xi) The slide chairs & bearing plates will remain on same position for both LH/RH T.O
15		LH: 04929:01 RH:04930:01			05493:02	05408:01 05409:02	Plate Screw RT-3912&3913:

							04
16		LH: 04924:01 RH:04925:01			04198:02	05485:02 05408:01	Plate Screw RT- 3912&3913:0 4
17		LH: 04929:01 RH:04930:01			05493:02	05080:01 05491:01	
18		LH: 04924:01 RH:04925:01			04198:02	05485:02	
19		LH: 04929:01 RH:04930:01			05493:02	05080:01 05491:01	
20		LH: 04924:01 RH:04925:01			04198:02	05485:02	
21		LH: 04929:01 RH:04930:01			05493:02	05080:01 05491:01	
22		LH: 04926:01 RH:04927:01			04198:02	05486:02	
23		LH: 04931:01 RH:04932:01			05493:02	05081:01 05492:01	
24		LH: 04926:01 RH:04927:01			04198:02	05486:02	
25		LH: 04926:01 RH:04927:01			04198:02	05486:02	
26			LH: 05026:01 RH:05027:01		04199:02 04200:02	05082:02	
27			LH: 05028:01 RH:05029:01		04199:02 04200:02	05082:02	
28				LH:05030:01 RH:05031:01	04811:04	05084:02	

SLEEPER WISE FITTINGS IN LEAD FOR 1 IN 12 555 R CANTED TURNOUT

Sleeper No.	Description of fittings with Drawing No & Quantity						
	Combined Bearing plate	Special Bearing plate			GRSP 10mm	GRSP 06mm	Remarks
		On LH side Track	On RH side Track	In Centre below VEE			
29	LH:05032:01 RH:05033:01				04811:04	05085:02	
30	LH:05034:01 RH:05035:01				04811:04	05086:02	i) All bearing plates shall be fixed on PSC sleepers by plate screws RT-3912 and 3913
31	LH:05036:01 RH:05037:01				04811:04	05087:02	
32	LH:05038:01 RH:05039:01				04811:04	05088:02	
33	LH:05040:01 RH:05041:01				04811:04	05089:02	ii) All plate screws shall be provided with single coil spring washer (SCSW) T-10773
34	LH:05042:01 RH:05043:01				04811:04	05090:02	
35	LH:05044:01 RH:05045:01				04811:04	05091:02	
36	LH:05046:01 RH:05047:01				04918 :04	005092:02	iii) ERC MK-V(RT-919 shall be fixed along with liners T-8222 and 8223
37	LH:05048:01 RH:05049:01				04918 :04	05093:02	
38	LH:05050:01 RH:05051:01				04918 :04	05094:02	
39		05476:02	05477:02		04918 :04	05488:04	iv) Combined bearing plate at sleeper 29 to 38 on both side
40		05476:02	05477:02		04918 :04	05488:04	
41		05476:02	05477:02		04918 :04	05488:04	
42		05476:02	05477:02		04918 :04	05488:04	v) Special bearing plate below VEE shall be fixed from sleeper 63 to 66 vi) Location of bearing plate shown will remain on same location for both LH/RH T.O.
43		05476:02	05477:02		04918 :04	05488:04	

44		05476:02	05477:02		04918 :04	05488:04	
45		05476:02	05477:02		04918 :04	05488:04	
46		05476:02	05477:02		04918 :04	05488:04	
47		05476:02	05477:02		04918 :04	05488:04	
48		05478:02	05479:02		04918 :04	05489:04	
49		05478:02	05479:02		04918 :04	05489:04	
50		05478:02	05479:02		04918 :04	05489:04	
51		05478:02	05479:02		04918 :04	05489:04	Location of bearing plate shown will remain same for both LH/RH T.O.
52		05478:02	05479:02		04918 :04	05489:04	
53		05478:02	05479:02		04918 :04	05489:04	
54		05478:02	05479:02		04918 :04	05489:04	
55		05478:02	05479:02		04918 :04	05489:04	
56		05478:02	05479:02		04918 :04	05489:04	
57		05478:02	05479:02		04918 :04	05489:04	
58		05478:02	05479:02		04918 :04	05489:04	
59		05478:02	05479:02		04918 :04	05489:04	
60		05478:02	05479:02		04918 :04	05489:04	
61		05478:02	05479:02		04918 :04	05489:04	
62		05478:02	05479:02		04918 :04	05489:04	
63		05478:01	05479:01	05503:01	04918 :04	05489:02 05504:02	
64		05478:01	05479:01	05052:01	04918 :04	05489:02 05095:01	
65		05478:01	05479:01	05053:01	04918 :02 04811:02	05489:02 05096:01	
66		05478:01	05479:01	05054:01	04918 :02 04811:02	05489:02 05097:01	

SLEEPER WISE FITTINGS IN CROSSING FOR 1 IN 12 555 R CANTED TURNOUT									
Sleeper No.	Special Bearing Plates					CRSP 10mm	GRSP 06mm	K-Type clip	Remarks
	Crossing B.P	Standard B.P	Special B.P	Check Rail B.P on RH XING	Check Rail B.P on LH XING				
67	5055-01no		LH:5480-01 RH:5481-01	LH:5132-01	5133-01 no	4811-02nos & 4918-02nos	5490-02nos,5022-01no,5098= 01no		i)Plate screws RT-3912/3913 shall be fixed at end of B.P having 20/28.5mm thickness respectively
68	5056-01no		LH:5480-01 RH:5481-01	LH:5132-01 RH:5133-01	5132-01 no & 5133-01 no	4198-02nos & 4918-02nos	5490-02nos,5022-02nos,5099= 01no	5177-01no with one T-head bolt	
69	5057-01no		LH:5480-01 RH:5481-01	LH:5132-01 RH:5133-01	5132-01 no & 5133-01 no	4198-02nos & 4918-02nos	5490-02nos,5022-02nos,5100= 01no	5178-01no with one T-head bolt	ii)GFN liners RT-8222/8223 shall be fixed with ERC Mk-V at the end of B.P having 20/28.5 mm thickness respectively
70	5058-01no		LH:5480-01 RH:5481-01	LH:5132-01 RH:5133-01	5132-01 no & 5133-01 no	4918-02nos & 5154-01no	5490-02nos,5022-02nos,5487= 01no		
71	5058-01no		LH:5480-01 RH:5481-01	LH:5132-01 RH:5133-01	5132-01 no & 5133-01 no	4918-02nos & 5154-01no	5490-02nos,5022-02nos,5487= 01no		iii)All plate screws and bolts shall be fitted with Single coil spring washers T-10773
72	5058-01no		LH:5480-01 RH:5481-01	LH:5132-01 RH:5133-01	5132-01 no & 5133-01 no	4918-02nos & 5154-01no	5490-02nos,5022-02nos,5487= 01no		iv) Check rails shall be fixed with square head M24×90 bolt with nut
73	5058-01no		LH:5480-01 RH:5481-01	LH:5132-01 RH:5133-01	5132-01 no & 5133-01 no	4918-02nos & 5154-01no	5490-02nos,5022-02nos,5487= 01no		v)Top of check rails shall be raised by 20mm from running rail

74	5059-01no		LH:5480-01 RH:5481-01	LH:5132-01 RH:5133-01	5132-01 no & 5133-01 no	4918-02nos & 5155-01no	5490-02nos,5022- 02nos,5111-01no		vi) 10mm thick CGRSP shall be used above B.P Where as 6mm GRSP shall be used below B.P
75	5060-01no		LH:5480-01 RH:5481-01	LH:5132-01 RH:5133-01	5132-01 no & 5133-01 no	4918-02nos & 5156-01no	5490-02nos,5022- 02nos,5112-01no		
76	5061-01no		LH:5480-01 RH:5481-01	LH:5132-01	5133-01 no	4918-02nos & 5157-01no	5490-02nos,5022- 02nos,5113-01no		vii) Longer check rail shall be fixed on Straight side and shorter check rail on curved side of T.O
77	5062-01no		LH:5480-01 RH:5481-01			4918-02nos & 5158-01no	5490-02nos, 5083-01no		
78	05063-01 no		LH:5480-01 RH:5481-01			4918-02nos & 4198-01no	5490-02nos, 5101-01no	5179-01no with one T-head bolt	viii) Location of bearing plate shown will remain same for both LH/RH T.O.
79	05064-01 no		LH:5480-01 RH:5481-01			4918-02nos & 4198-01no	5490-02nos, 5102-01no	5180-01no with one T-head bolt	
80	05065-01 no		LH:5480-01 RH:5481-01			4918-02nos & 4811-02 nos	5490-02nos, 5103-01no		
81	05066-01 no		LH:5480-01 RH:5481-01			4918-02nos & 4811-02 nos	5490-02nos, 5104-01no		
82	05067-01 no		LH:5480-01 RH:5481-01			4918-02nos & 4811-02 nos	5490-02nos, 5105-01no		
83	05068-01 no		LH:5480-01 RH:5481-01			4918-04nos	5490-02nos, 5106-01no		
84			LH:5480-01 RH:5481-01			4918-04nos	5490-04nos		
85			LH:5480-02 RH:5481-02			4918-04nos	5490-04nos		On LH Track special B.P 5480 & on RH Track Special BP 5481 shall be fixed & location will remain same for both LH/RH Turnout
86			LH:5480-02 RH:5481-02			4918-04nos	5490-04nos		
87			LH:5480-02 RH:5481-02			4918-04nos	5490-04nos		

88			LH:5480-02 RH:5481-02			4918-04nos	5490-04nos		
89			LH:5480-02 RH:5481-02			4918-04nos	5490-04nos		
90			LH:5480-02 RH:5481-02			4918-04nos	5490-04nos		
91			LH:5480-02 RH:5481-02			4918-04nos	5490-04nos		
92			LH:5480-02 RH:5481-02			4918-04nos	5490-04nos		
93		5473-4 nos				4918-04nos	5483-04nos		
94		5473-4 nos				4918-04nos	5483-04nos		

S.NO	CHECK SHEET OF SWITCH ASSEMBLY(O4756-04) FOR 1 IN 12 555R CANTED TURNOUT						
	DISCRIPTION OF ITEMS		DESIGNED VALUE mm	MEASURED VALUES ON ASSEMBLED/LAID SET			
				T.O NO.... LH/RH		T.O NO... LH/RH	
				LH	RH	LH	RH
1	Sleeper spacing in switch portion	See Separate sheet					
2	Distance between Gauge faces of outer rails	See Separate sheet					
3	Distance SRJ - ATS	T.P & SRJ at same point	745.5±6				
4	Distance ATS- 72 MM Head (JOH)	Curved/Straight (End of side Machining)	8013/8000				
5	Location of Anticreep Device	Between SL no 26-27	In Centre				
6	Offset at End of Stock rail (By scale)	M/L side	257±2				
		T.O Side	257.7±2				
7	Straightness of Straight Stock Rail	By Thread & scale	± 01				
8	Versine of Curved Tongue rail at (4042.5, 8085, 12127.5) ± 01 (By thread & scale)	V1	44.1				
		V2	58.8				
		V3	44.1				
9	Straightness of Straight Tongue Rail	By Thread & scale	± 01				
10	Versine of Curved stock rail at (4225, 8450, 12675) ± 01 (By thread & scale)	V1	48.3				
		V2	64.4				
		V3	48.3				

11	Switch set for Main Line						
	i) Throw (If Point M/C fixed) ii) Clearance on Open side/Gap on closed side (When point M/C is fixed)	At ATS At JOH (with effective back drive)	160 60/0				
	iii) Matching of Stock & Tongue/ Rail	TOE to JOH	within 01				
12	Switch Set for Turnout						
	i) Throw(If Point M/C fixed) ii) Clearance on Open side/Gap on closed side (When point M/C is fixed)	At ATS At JOH (with effective back drive)	160 60/0				
	iii) Matching of Stock & Tongue/ Rail	TOE to JOH	within 01				
13	Distance between Gauge faces of Stock rail at JOH	By Tape	1748				
14	Track Gauge	At alternate sleepers From SL no 01 to 28	Main Line 1676 +2,-0 Turnout 1676 +2, -0				
15	Cross Level	At alternate sleepers From SL no 1 to 28					
16	Piroll with Double Roller	AT SL NO.2,5,8,11, 13,15	OK/not OK				
17	Piroll with Single Roller	AT SL NO 17,19	OK/not OK				
18	Piroll with Spring loaded Roller	AT SL NO 21,23	OK/not OK				
19	M.S Plate	At SL NO 3,16	OK/not OK				
20	Insulated Tie Plate	AT SL NO 2	OK/not OK				
21	Fittings in switch assembly	Complete fittings as per drawing	OK/not OK				
22	Fitment of Back drive	At SL NO 2,3,8,13 & 14	On opposite side of point M/C on Base plates				

23	Fitment of two point machines	Between SL NO:2-3					
24	Effectiveness of Anticreep system	Equal side gaps/tight bolts	OK/Not OK				
25	Versine of curved tongue rail (To match with curvature of lead rail) Starting from SRJ	At 6 m chord and 3 M interval	8				
26	Squareness of stock rails	At SRJ	± 3				
27	Gap between slide block & T/rail web	At each Slide block	within 01				
28	Tightness of bolts		OK/Not OK				
29	Tightness of ERC		OK/Not OK				
30	Tightness of wedge/Leaf springs	On all slide chairs	OK/Not OK				
31	Tightness of plate screws/other fittings	On all chairs	OK/Not OK				
32	Effectiveness of 2nd point M/C		OK/Not OK				
33	Fitment of all Chairs /fittings/GRSP/CGSP on position	As per relevant drawings	OK/Not OK				
34	Deficiencies						
35	Suggestions/ Remarks						

CHECK SHEET OF LEAD ASSEMBLY FOR 1 IN 12 555R CANTED TURNOUT

S.NO	DESCRIPTION OF ITEMS		DESIGNED VALUE mm	MEASURED VALUES ON ASSEMBLED/LAID SET			
				T.O NO...	LH/RH	T.O NO...	LH/RH
1	Sleeper spacing	Sleeper no:28 to 66	(See separate sheet)				
2	Distance between gauge faces of outer rails	Sleeper no:28 to 66	(See separate sheet)				
3	Offset in Lead at Sleeper Nos	Gauge face of Straight rail to GF of curved rail					
	29	268					
	32	327					
	34	370					
	37	439					
	39	488					
	42	568					
	44	624					
	47	713					
	49	775					
	52	874					
	54	943					
	57	1052					
	59	1128					
	62	1246					
	64	1328					
	67	1456					
4	Track gauge	At alternate sleepers starting from SL no: 29 to 67	Main Line 1676+2,-0 Turnout 1676+2,-0				
5	Cross level	At Alternate sleepers					
6	Combined bearing plate	SL no: 29 to 38	Fixation O.K/Not O.K				
7	Single foot bearing plate	On SL no: 39 -47 LH/RH side On SL no: 48-66 On LH/RH side	05476/05477 & 05478/05479 (one on each rail seat)				

8	Curvature of lead	At 06M chord & 03M intervals (In continuation of switch curve)	08 mm		
9	Fitment/Tightness of Plate screws		OK/Not OK		
10	Tightness of ERC		OK/Not OK		
11	Fitment on all Chairs /fittings/GRSP/CGR SP on position	As per relevant drawings			
12	Deficiencies				
13	Suggestions/ Remarks				

S.NO	CHECK SHEET OF WELDABLE CROSSING ASSEMBLY FOR 1 IN 12 555R CANTED TURNOUT					
	DISCRIPTION OF ITEMS	DESIGNED VALUE mm	MEASURED VALUES ON ASEMBLED/LAID SET			
			T.O No Xing SLNO	T.O No Xing SLNO	T.O No Xing SLNO	T.O No Xing SLNO
			LH/RH	LH/RH	LH/RH	LH/RH
1	Sleeper spacing	See separate sheet				
2	Nose Clearance LH/RH	44 +0,-2				
3	Throat Clearance	66 +2-0				
4	Wing rail Clearance LH/RH	66 +3,-0				
5	Nose Slope i) 4 mm down (At 385 from TNC) ii) 8mm down(At 116 from TNC)	i) 4 ±0.5 ii)8 +2,-0.5				
6	Curved length of Xing (Upto 2090 from heel end of Xing)	6338				
7	Curvature of Xing at 6m chord at 3 m interval (In continuation to lead rail curve)	08 (Up to 2088mm from heel end of Xing)				
8	Straightness of Xing ±1	In Full length on Str. Side				
9	Check rail clearance LH/RH	44				
10	Raising of check rail LH/RH (Difference between top level of running & check rails)	20				
11	Track gauge at alternate sleepers	Main Line 1676+2,-0 Turnout 1676+2,-0				

12	Cross Level	At alternate sleeper				
13	Fitment of Bearing plates with proper GRSP/CGRSP	At proper location as per dwg.				
14	Fitment/Tightening of Plate screw/ERC/Bolts of check rail chairs	As per Dwg				
15	Condition of top surface of Xing	Should be smooth & free from pitting/Chipping				
16	Deficiencies					
17	Suggetions/Remarks					

CHECK SHEET FOR 1:12 555R PSC SWITCH SLEEPERS

SWITCH PORTION SLEEPER NO: 01 TO 28

SLEEPER NO	LENGTH mm	DISTANCE BETWEEN RUNNING FACES OF OUTER RAILS mm	SPACING BETWEEN SLEEPERS					REMARKS
			BETWEEN SLEEPER NO	SPACING ON OUTER STRAIGHT SIDE mm	MEASURED VALUES	SPACING ON INNER CURVED SIDE mm	MEASURED VALUES	
1	2750	1676.1	SRJ-01	300		300		Sleeper No: 01 TO 28 are laid in Fan Shape
2	4160	1676.7	1 to 2	600		598		
3	4160	1678.4	2 to 3	745		745		
4	2750	1680.5	3 to 4	600		599		
5	2750	1683.3	4 to 5	600		599		
6	2750	1686.7	5 to 6	600		599		
7	2750	1690.7	6 to 7	600		599		
8	3585	1695.4	7 to 8	600		599		Sleeper spacing are measured on outer foot of straight and curved rails
9	2760	1700.8	8 to 9	600		599		
10	2760	1706.8	9 to 10	600		599		
11	2760	1713.4	10 to 11	600		599		
12	2780	1720.7	11 to 12	600		599		
13	3620	1728.6	12 to 13	600		599		
14	3630	1737.2	13 - 14	600		599		
15	2800	1745.7	14 - 15	600		599		
JOH	0	1748						
16	2820	1757	15 - 16	600		599		
17	2820	1766.8	16 - 17	600		599		
18	2840	1778	17 - 18	600		599		

19	2840	1789.8	18 - 19	600		599		
20	2860	1802.3	19-20	600		599		
21	2880	1815.4	20 -21	600		599		
22	2880	1829.2	21 - 22	600		599		
23	2900	1843.6	22 - 23	600		599		
24	2920	1858.6	23 - 24	600		599		
25	2920	1874.4	24 - 25	600		599		
26	2940	1890.7	25 - 26	600		599		
27	2960	1907.7	26 - 27	600		599		
28	2980	1925.4	27-28	600		599		

CHECK SHEET FOR 1:12 555R PSC LEAD SLEEPERS OF CANTED TURNOUT

	LEAD PORTION SLEEPER NO: 29 TO 66							REMARKS
SLEEPER NO	LENGTH mm	DISTANCE BETWEEN RUNNING FACES OF OUTER RAILS mm	SPACING BETWEEN SLEEPERS					
			BETWEEN SLEEPER NO	SPACING ON OUTER STRAIGHT SIDE mm	MEASURED VALUES	SPACING ON INNER CURVED SIDE mm	MEASURED VALUES	
29	3000	1943.7	28-29	600		599		Sleeper No 28 to 66 in lead portion are laid in Fan Shape
30	3020	1962.6	29-30	600		599		
31	3040	1982.2	30-31	600		598		
32	3060	2002.6	31-32	605		604		
33	3080	2023.7	32-33	605		604		
34	3100	2045.4	33-34	605		604		
35	3120	2067.8	34-35	605		604		
36	3140	2090.9	35-36	605		604		Sleeper spacing are measured on outer foot of straight and curved rails
37	3160	2114.6	36-37	605		604		
38	3200	2138.9	37-38	605		603		
39	3220	2164	38-39	605		604		
40	3240	2189.6	39-40	605		604		
41	3280	2216	40-41	605		604		
42	3300	2243	41-42	605		603		
43	3320	2270.6	42-43	605		604		
44	3360	2298.9	43-44	605		604		
45	3380	2327.9	44-45	605		603		
46	3420	2357.5	45-46	605		604		
47	3440	2387.8	46-47	605		604		
48	3480	2418.8	47-48	605		604		
49	3500	2450.4	48-49	605		603		
50	3540	2482.6	49-50	605		604		
51	3580	2515.5	50-51	605		603		
52	3600	2549.1	51-52	605		604		
53	3640	2583.3	52-53	605		604		
54	3680	2618.2	53-54	605		603		
55	3700	2653.8	54-55	605		604		
56	3740	2690	55-56	605		603		

57	3780	2726.8	56-57	605		603		
58	3820	2764.4	57-58	605		604		
59	3860	2802.5	58-59	605		604		
60	3900	2841.4	59-60	605		603		
61	3940	2880.8	60-61	605		603		
62	3980	2920.7	61-62	600		599		
63	4020	2961.1	62-63	600		598		
64	4060	3002.2	63-64	600		598		
65	4100	3044	64-65	600		599		
66	4140	3086.4	65-66	600		598		

CROSSING PORTION FROM SLEEPER NO: 67 TO 94

SLEEPER NO	LENGTH mm	DISTANCE BETWEEN RUNNING FACES OF OUTER RAILS mm	SPACING BETWEEN SLEEPERS					REMARKS
			BETWEEN SLEEPER NO	SPACING ON OUTER STRAIGHT SIDE mm	MEASURED VALUES	SPACING ON INNER CURVED SIDE mm	MEASURED VALUES	
67	4180	3129.4	66-67	600		598		Sleeper No 66 TO 77 are laid in Fan Shape
68	4220	3173.1	67-68	600		599		
69	4280	3217.4	68-69	600		598		
70	4320	3262.4	69-70	600		598		
71	4360	3308	70-71	600		598		
72	4400	3354.3	71-72	600		598		
73	4460	3401.2	72-73	600		599		Sleeper spacing are measured on outer foot of straight and curved rails
74	4500	3448.7	73-74	600		598		
75	4560	3496.9	74-75	600		598		
76	4600	3545.8	75-76	600		598		
77	4660	3595.3	76-77	600		598		
78	4700	3645.2	77-78	600		600		
79	4760	3695	78-79	600		600		
80	4800	3744.9	79-80	600		600		
81	4840	3794.8	80-81	600		600		
82	4900	3844.6	81-82	600		600		
83	4940	3894.5	82-83	600		600		
84	5000	3944.4	83-84	600		600		
85	5040	3994.2	84-85	600		600		
86	5100	4044.1	85-86	600		600		
87	5140	4094	86-87	600		600		
88	5200	4143.9	87-88	600		600		
89	5240	4193.7	88-89	600		600		
90	5300	4243.6	89-90	600		600		
91	5340	4293.5	90-91	600		600		
92	5400	4343.3	91-92	600		600		
93	2710	4393.3	92-93	600		600		
94	2710	4443.2	93-94	600		600		

6. INSTALLATION OF TURNOUT ITEMS

Use professional and suitable devices for handling. The lifting devices and equipment for transportation must be suitable for handling the delivered goods. For instance, the lifting force of the cranes must correspond to the weight of the packages; the yards must be big enough, plain, drained and easily accessible. Weights and lengths of the packages are of particular importance.

For long packages, provide an adequate number of lifting points. Do not drop the cargo or bump against something because this could lead to displacements of tied and preassembled parts and consequently to damage. Apart from that, do not lift long parts like crossings, rails, stock rails, switches at only one end or unsymmetrically (sagging).

For lifting of turnout's main parts, cranes with traverse (cross beams) are used. The parts have to be hoisted by load belts and lifting hooks. Use of chains is prohibited.

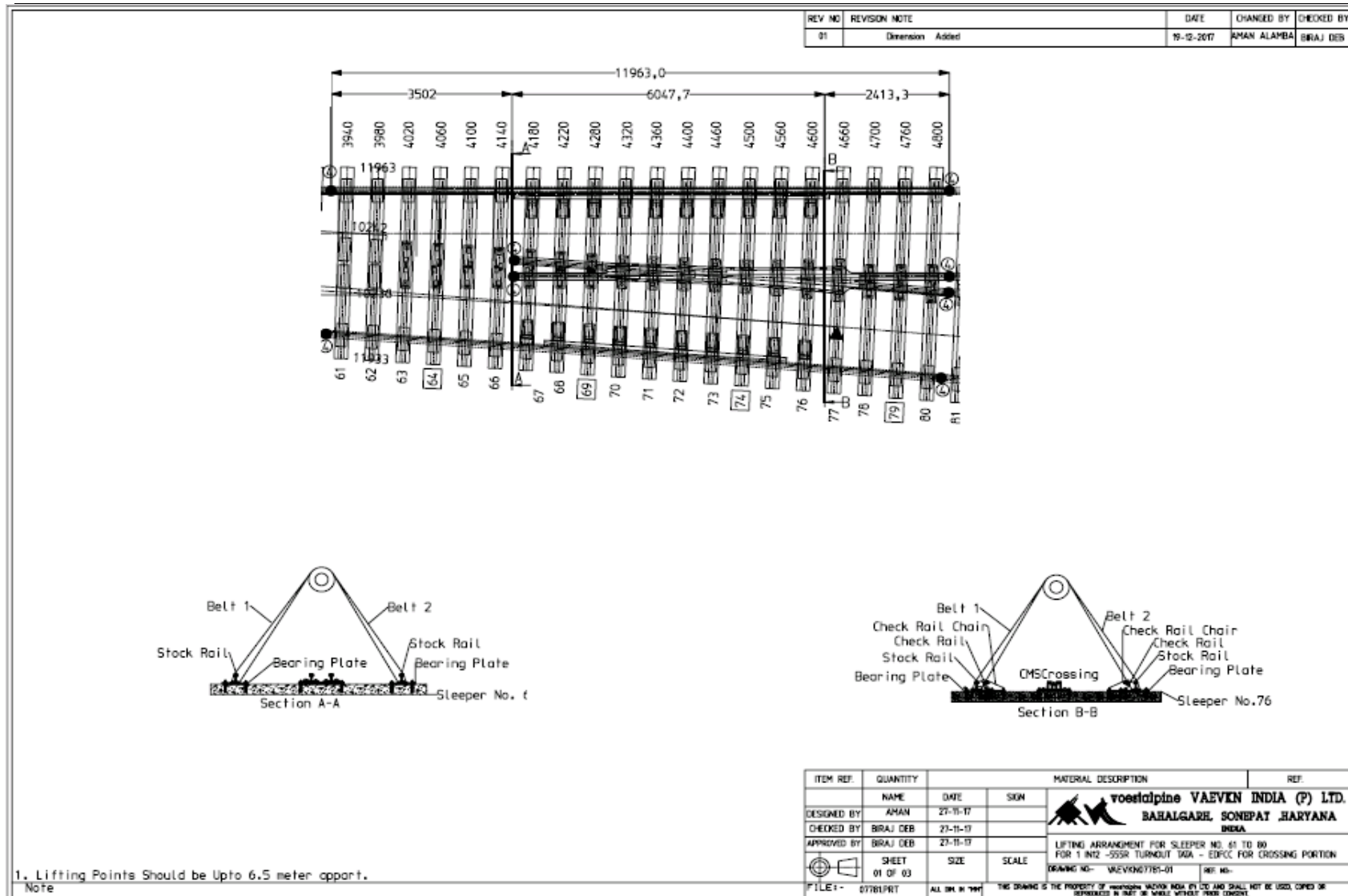
The turnout parts should be lifted in a way that as little stress as possible is induced into turnout components (i.e., an adequate number of lifting points must be provided for long parts). Special care must be taken when handling rails with glued insulated joints.

Lifting points should be upto 6.5m apart.

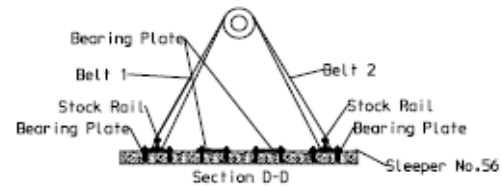
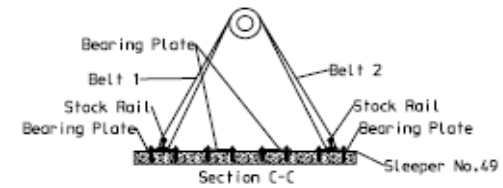
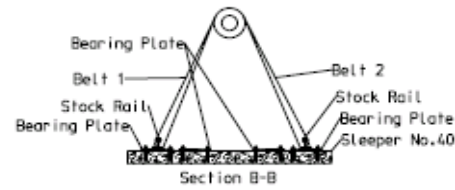
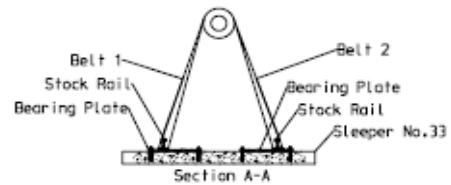
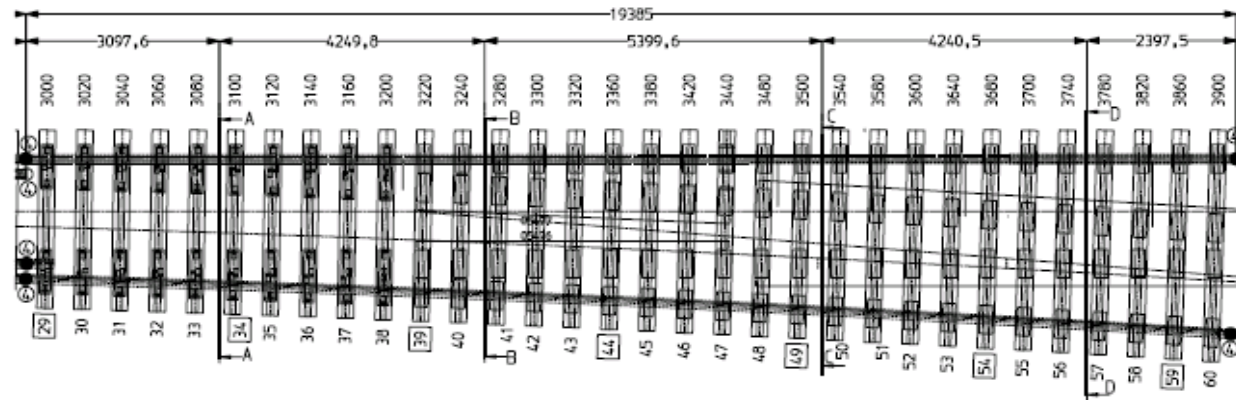
Examples of lifting of rails and switch device:





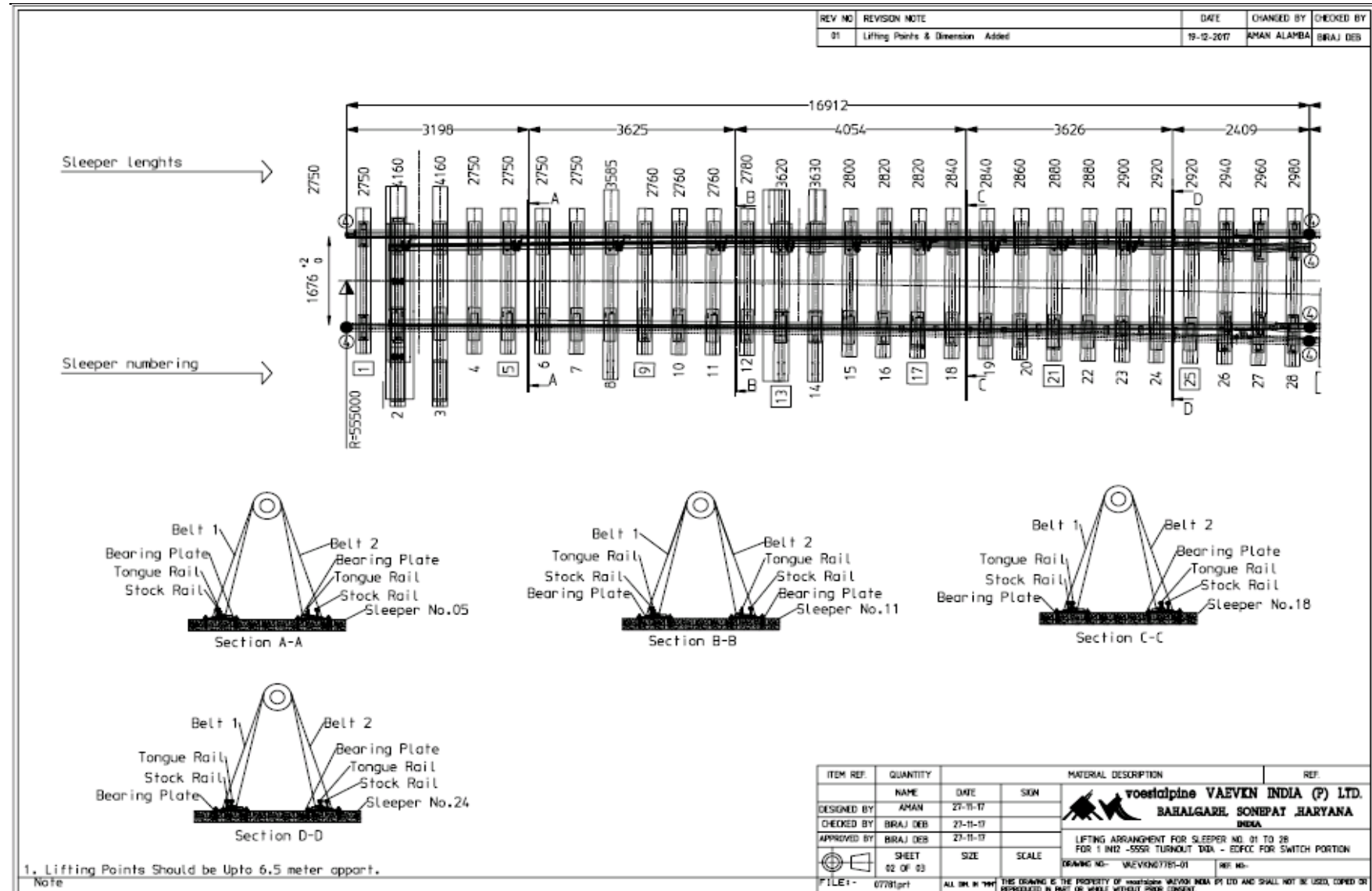


REV. NO.	REVISION NOTE	DATE	CHANGED BY	CHECKED BY
01	Lifting Points & Dimension Added	19-12-2017	AMAN ALAMBA	BIRAJ DES



1. Lifting Points Should be Upto 6.5 meter appart.
NOTE

ITEM REF.	QUANTITY	DATE	SIGN	MATERIAL DESCRIPTION	REF.
DESIGNED BY	AMAN	27-11-17		voestalpine VAIVEN INDIA (P) LTD.	
CHECKED BY	BIRAJ DES	27-11-17		BAHALGARH, SONEPAT, HARYANA	
APPROVED BY	BIRAJ DES	27-11-17		INDIA	
SHEET	03 OF 03	SIZE	SCALE	LIFTING ARRANGEMENT FOR SLEEPER NO. 29 TO 60 FOR 1 IN 2 -SSR TURNOUT JAWA - EDICE FOR LEAD PORTION	
FILE:-	07781prt	ALL DIM IN MM		DRAWING NO:- VAIVEN07785-01	REV. NO:-
THIS DRAWING IS THE PROPERTY OF voestalpine VAIVEN INDIA (P) LTD AND SHALL NOT BE USED, COPIED OR REPRODUCED IN PART OR WHOLE WITHOUT PRIOR CONSENT					



6.1 Step – By – Step: Installation Procedure for 1in12 Turnout

STEP-1: 96 Sleepers to be placed on the surface.

STEP-2: Sleepers serial no. 1 to 96 to be placed serially starting from serial no. 1.

STEP-3: Sleeper serial no. should always be kept on right hand side in LH as well as RH T/O when viewing from facing direction.

STEP-4: Sleepers should be placed according to sleeper spacing distance shown in the drawing.

STEP-5: For LH T/O right hand end of the sleepers should be kept aligned and for RH T/O left hand end of sleepers should be placed aligned.

STEP-6: GRSP should be placed on the sleepers as per table given in the drawing no. VAEVKN04743-04, VAEVKN04756-04 and VAEVKN05107-03.

STEP-7: Bearing plates also to be placed at the location shown in the above drawings on the GRSP.

NOTE:

(a) For LH & RH T/O same set of sleepers can be used.

(b) Location of bearing plates as well as rubber pad is shown in the drawing no. VAEVKN04743-04 which is a RH T/O.

For LH T/O the location of bearing plates & 6mm GRSPs will be same as shown in the RH T/O.

For further clarification please see note no. 5 of drawing no. VAEVKN04743-04.

STEP-8: Bearing plates & slide chairs to be fixed with sleepers with plate screws & single coil spring washer as given in the table of the above drawings.

STEP-9: For RH T/O left hand stock rail to be dismantled from the assembled switch and should be placed on the slide chairs.

STEP-10: The lead rail behind the straight stock rail according to the specific length as shown in the drawing should be placed behind the left hand stock rail.

STEP-11: Straight lead rail should be placed upto Serial no. 96 as shown in the drawing.

STEP-12: Stock rail & the straight lead rail should be fastened with liner, ERC, leaf spring and wedge as shown in the drawing.

STEP-13: The straight rail starting from sleeper no. 1 to 96 should be aligned within $\pm 2\text{mm}$ with the help of a versine kit.

STEP-14: LH tongue rail should be placed with the stock rail.

STEP-15: Tongue rail should be placed in such a way that the Anticreep pin already fixed with the tongue rail is at the centre of the Anticreep fork leaving a gap of 7mm approximately on the either side of the fork.

- STEP-16:** The tongue rail should be fastened with liner and ERC at sleeper no. 26, 27 & 28.
- STEP-17:** Curved lead rail behind the curved tongue rail to be placed upto sleeper no. 66.
- STEP-18:** Offset in between the straight and curved running edge upto sleeper no. 66 should be maintained as per the chart (running edge offset) given in the drawing.
- STEP-19:** Curved lead rail to be fastened with baseplate with ERC and liners.
- STEP-20:** Curved stock rail to be placed on the opposite side of the straight stock rail.
- STEP-21:** Sleepers are to be arranged in a fan shaped layout. To achieve the fan shaped layout, the sleepers are to be placed at a sleeper spacing shown in the drawing no. VAEVKN04743-04. Sleeper spacing should be measured at the field side foot of the curved as well as the straight outer rails of the turnout.
- STEP-22:** Track gauge to be checked from the curved tongue rail to the curved stock rail for every two alternative sleepers.
- STEP-23:** Curved lead rail to be placed behind the curved stock rail upto sleeper no. 94 and should be checked for track gauge and if it is found satisfactory it should be fastened with proper fastening system.
- STEP-24:** Tongue rail to be placed beside the stock rail and in such a way that the Anticreep pin already fixed with the tongue rail is at the centre of the Anticreep fork leaving a gap of 7mm approximately on the either side of the fork.
- STEP-25:** The tongue rail should be fastened with liner and ERC at sleeper no. 26, 27 & 28.
- STEP-26:** Straight lead rail behind the straight tongue rail to be placed upto sleeper no. 66.
- STEP-27:** Track gauge to be checked at lead area as well as at switch area.
- STEP-28:** Straight lead rail to be fastened with base plates, liners and ERC.
- STEP-29:** Crossing to be placed as shown in the drawing from sleeper no. 67 to 80, just behind the two inside lead rails.
- STEP-30:** Another two lead rails to be placed behind the crossing.
- STEP-31:** Track gauge to be checked on the either side of the crossing & lead rails and to be fastened with baseplates with ERC & Liners.
- STEP-32:** Check rail chairs are to be fastened from sleeper no 67 to 76.
- STEP-33:** Baseplate no VAEVKN05133 should always be fixed/fastened on the right hand side of the turnout whether it is a LH or RH T/O.
Baseplate no VAEVKN05132 should always be fixed/fastened on the left hand side of the turnout whether it is a LH or RH T/O.
- STEP-34:** Check rail should be placed as shown in the drawing to achieve a flange-way clearance 44mm $\pm$ 1mm.

7. TURNOUT DESCRIPTION

1.0	1 in 12 - 555R Ballasted Turnout	VAEVKN04743
1.1	Switch device	VAEVKN04756
1.2	Crossing	VAEVKN04745
1.3	Checkrails	VAEVKN05024 & VAEVKN05025

Part 1: Switch device

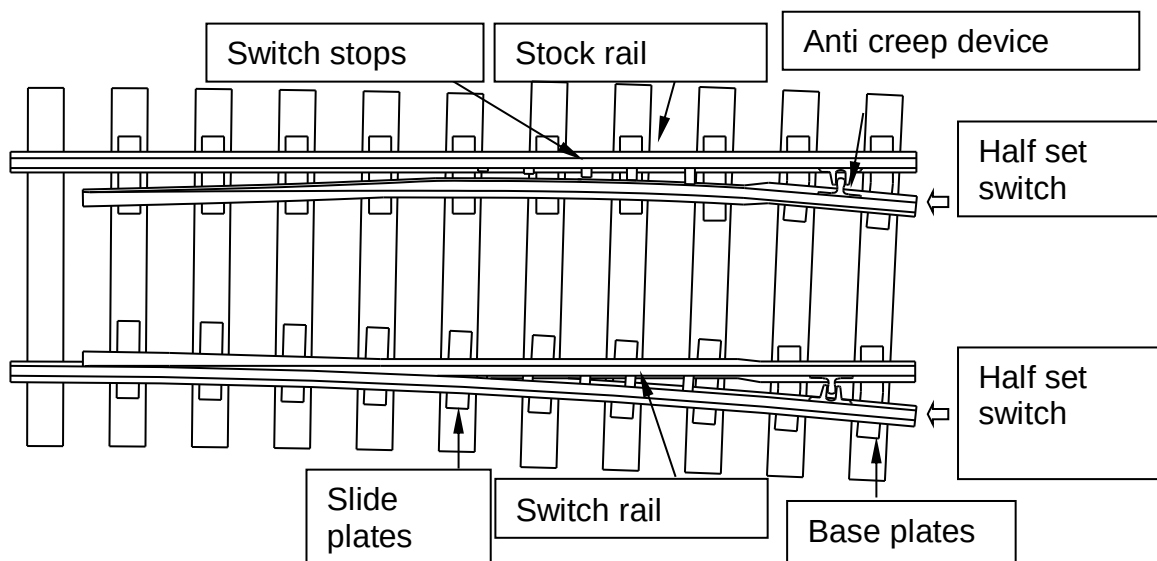


FIG. 1

Part 2: Intermediate rails

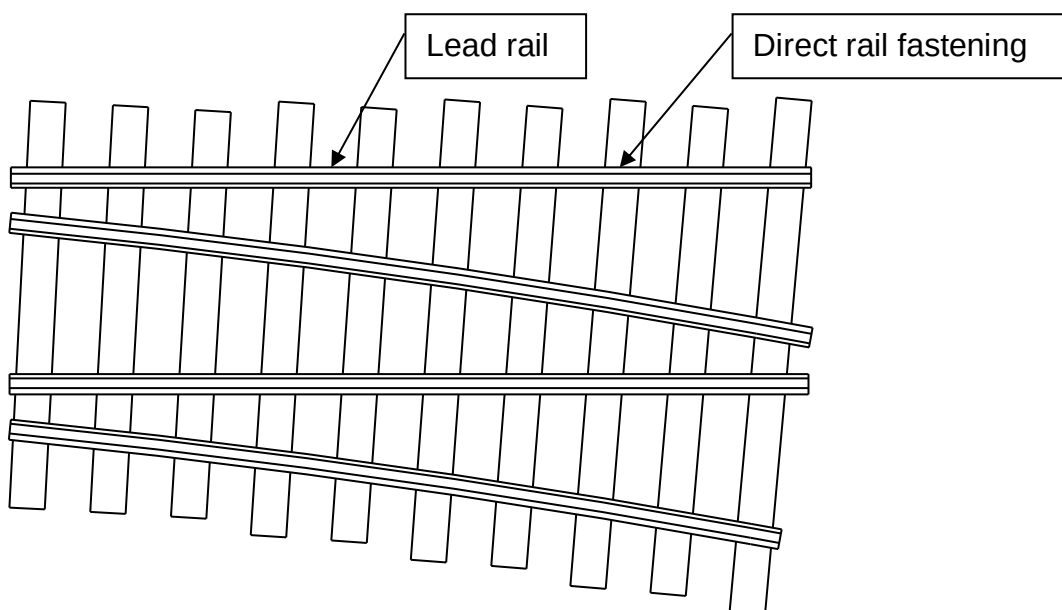


FIG. 2

Part 3: Crossing and check rail area

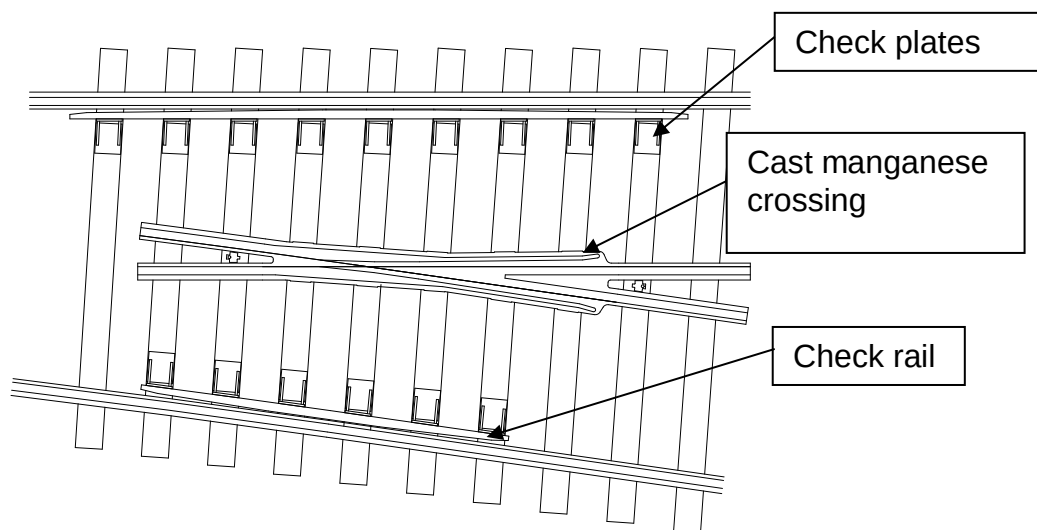


FIG. 3

PART - B

8. GENERAL RECOMMENDATIONS AND USE OF THIS HANDBOOK

The turnout has to be inspected in certain intervals to detect irregularities and to remove or repair them. That means that the turnout has to be inspected by two methods

- Visual inspection of the turnout and all the parts associated with it and manual dimensional check
- Check by a measuring tape

The interval of the preventive check is shown in “chapter 11– preventive maintenance”. Generally, these inspections have to be done in 2 or 6 month intervals. This manual will not deal with safety regulations that have to be fulfilled during the inspection, further it will not describe the details and types of measuring tapes that will be used, in this case internal EDFCC regulations have to be followed.

If failures will be detected, they have to be maintained according to the recommendations shown in column of maintenance action. Defects that cannot be removed by the preventive package of measures have to be determined according to “chapter 12- corrective maintenance”.

Additional activities regarding maintenance and inspection could become necessary if failure occurs in between inspection intervals. Irregularities during train passages, noticed by the train operator or increased setting forces could be main indicators for problems with one or more parts of the turnout. In such cases the turnouts have to be maintained and inspected immediately according to chapter 10.

Basically all components described in chapter 11 and 12 have to be inspected within required intervals, except the signalling equipment, for that the regulations from the signalling supplier have to be followed.

REPLACEMENT OF PARTS

It does not always make sense to replace single parts of turnouts (e.g. a single switch blade or a stock rail) as parts have to be fitted to each other. Due to this reason the turnout will be split in LRUs (Line Replacement Units)

Parts marked with „●“ are assemblies of LRU that contain all parts which are forming the LRU, parts marked with „○“ could be replaced as single unit but has to be fitted in some cases.

- Half set of switches
 - Switch stops
 - Fastening material for switch stops
 - Anti-creep device
 - Fastening material for anti-creep device
- Common /obtuse /acute cast manganese crossing
- Intermediate rails
- Base plates
- Slide plates
- PI rollers
- Mk-V Clips
- Leaf springs & Wedge
- GFN Liners

9. FAILURE DEFINITION

9.1.1 Rail crack/cross break

Cracks or cross breaks through the whole rail or cracks running from the railhead to the bottom.



9.1.2 Large surface defects

Large running surface defects on the railhead



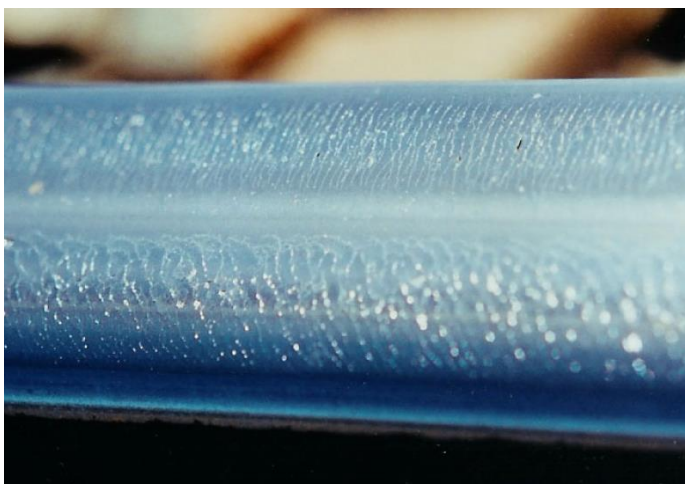
9.1.3 Cracks in the railhead:

Short cracks in the running surface, on the rail top as well as cracks close to the running edge and in the area of the head radius or the side of the railhead.



9.1.4 RCF of running surface

Typical rolling contact fatigue failures on the running surface.



9.1.5 Critical breakouts on the running edge

Critical breakouts close to the running edge that cannot be maintained by grinding of the railhead.



9.1.6 Small running edge defects

Smaller breakouts in the area of the running edge that does not influence the wheel rail contact behaviour and could be maintained or repaired by grinding.



9.1.7 Pollution of sliding chairs

Heavy polluted sliding chairs that hinder the blade movement.

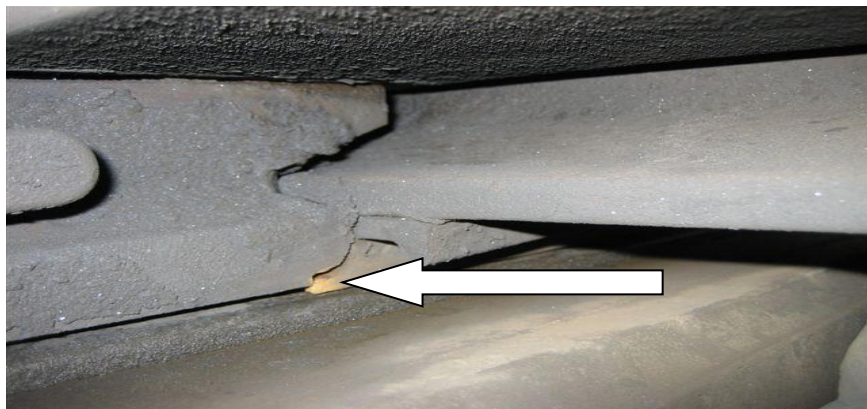
9.1.8 Heavy deformed plastic pads / breakage of plastic pads

Extremely deformed plastic pad and broken pads to be replaced.



9.1.9 Broken base / slide plates

Broken slide plates or base plates and cracks in plates.



10.1.2 Check „P1“ in wheel transition area

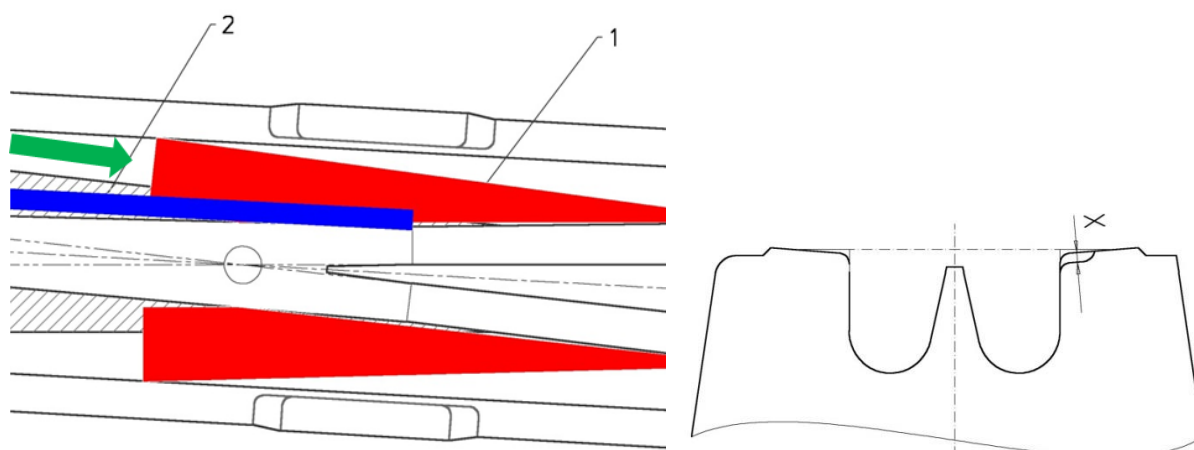
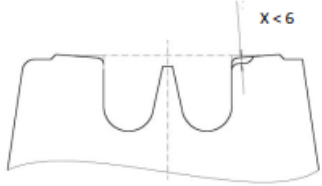
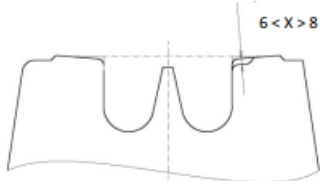
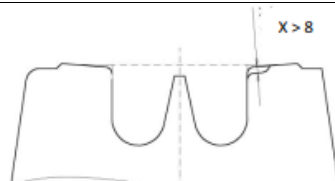


Fig. 2-Check of wing rail wear in wheel transition area

- 1) Zone of maximum wear
- 2) Straightedge

Equipment	Straightedge 1m, Feeler gauge			
Procedure	Place straightedge on wing rail and push it towards the wheel transition area. Measure the wing rail wear (X) using the feeler gauge.			
	Test Result	Operating condition		Measure
	$X < 6\text{mm}$	The crossing is still usable.	 GOOD	Not necessary
	$6\text{mm} < X < 8\text{mm}$	The crossing is usable, but in an advanced state of wear.	 MEDIUM	Schedule measure (Repair welding)
	$X > 8\text{mm}$	The crossing is not usable anymore.	 BAD	Define speed restriction and implement repair welding immediately

10.1.3 Check „P2 in crossing nose area

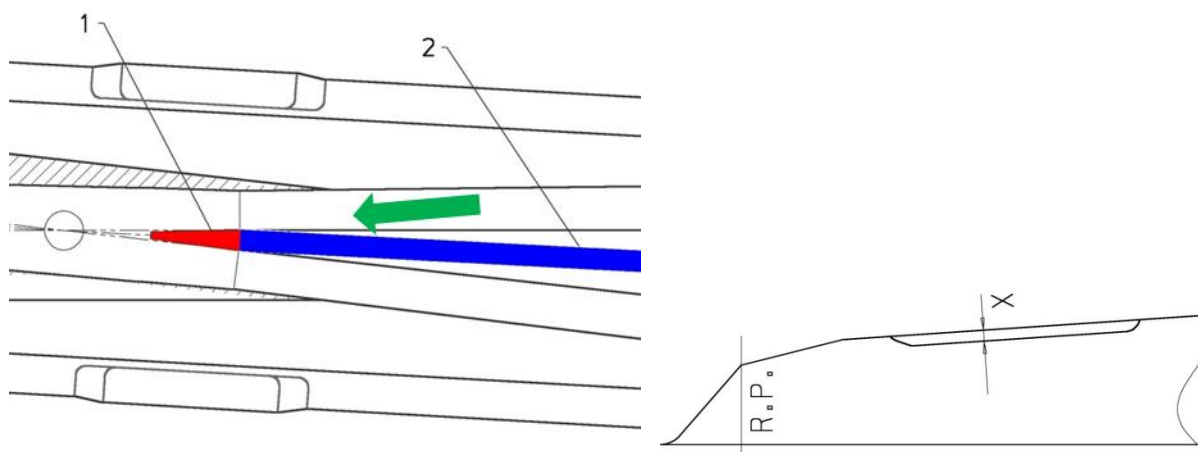


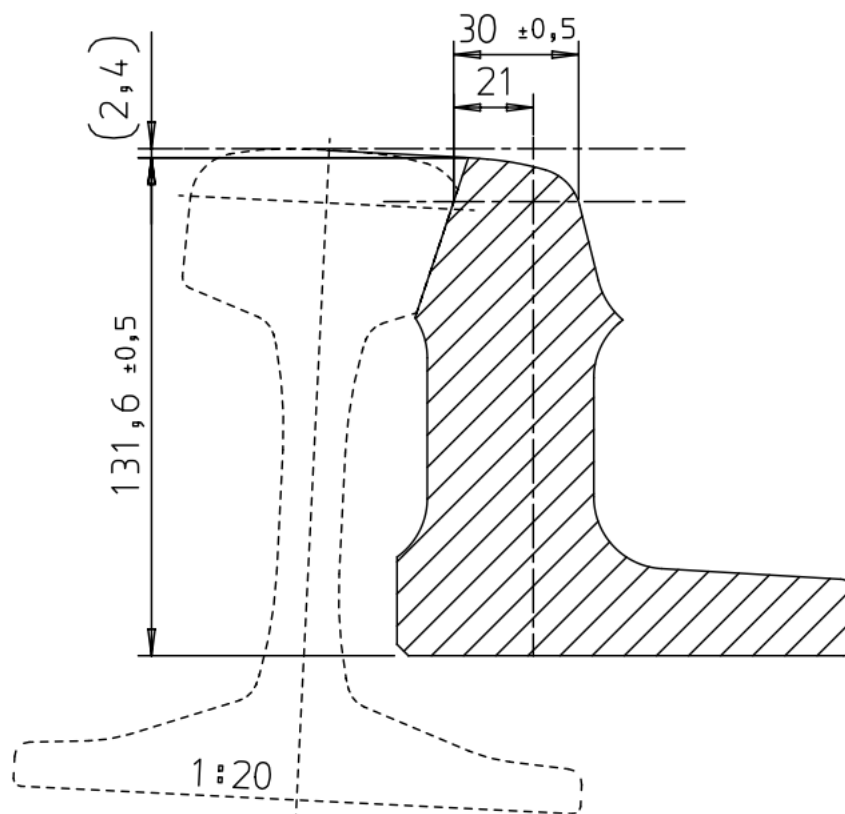
Fig. 3-Check of the nose topping in crossing nose area

- 1) Zone of maximum wear
- 2) Straightedge

Equipment	Straightedge 1m, Feeler gauge			
Procedure	Place straightedge on crossing nose and push it towards to the end of crossing nose. Measure the crossing nose wear (X) using the feeler gauge.			
	Test Result	Operating condition		Measure
	• $X < 3\text{mm}$	The crossing is still usable.		Not necessary
	• $3\text{mm} < X < 4\text{mm}$	The crossing is usable, but in an advanced state of wear.		Schedule measure (Repair welding)
	• $X > 4\text{mm}$	The crossing is not usable anymore.		Define speed restriction and implement repair welding immediately

10.2 Wear Limit of Switch

The aim of this guide is to define test applications, to check the wear on switch, which evolves during operation. The measures to be taken will be defined, depending on the degree of wear, so that derailment safety is kept on a tolerable level and switch can be replaced in time due to the objective state of wear. Wear is a normal phenomenon, which may occur especially in heavily loaded turnouts or turnouts with small radii. In the case of Switch, the location of 30mm head of Tongue rail from ATS is highly impacted.



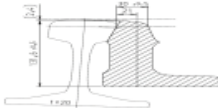
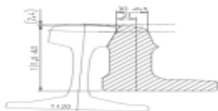
		voestalpine VAE VKN INDIA (P) LTD.			
PROJECT:- EDFCC - TATA				DATE :- 28-05-2020	
PARTICULARS OF TONGUE RAILS SHOWING LOCATION AND HEAD THICKNESS					
S.No.	Description	Drg. No. of Tongue Rail	Image	Location of 30mm Head From ATS	Location of JOH From ATS
1	Drg. No. of T/O-1 in 12-555-60E1_VAEVKN04743-04	Drg. No. of Straight Tongue Rail_VAEVKN04748-01		4727	8000
	Drg. No. of Switch Sub Assembly-1 in 12-555-60E1_VAEVKN04756-04	Drg. No. of Curved Tongue Rail_VAEVKN04747-02		4732	8013

Table: 1. Location of Checking the wear of Switch

10.2.1 Operating Condition of Switch Devices

The operating conditions described in the following chapters are classified into three categories in accordance with table below:

		
GOOD	MEDIUM	BAD
The half set of switches is safe against derailment and still usable.	The half set of switches can still be passed by trains but is in an advanced state of wear, schedule measure soon.	The half set of switches is not safe against derailment.

Further to the outcome of the tests, proceed in accordance with chapter **Error! Reference source not found.** When the operating status is “bad”, the turnout is not safe and must not be passed by trains anymore.

10.2.2 Vertical & Lateral wear of Switch

Procedure: The switch rail must rest on the slide plates properly. The turnout has to be set before the test and the switch must be left in the achieved position.			
Test Result	Operating condition		Measure
• Lateral wear < 4mm	The half set of switches is safe against derailment and still usable.	 GOOD	Not necessary
• Vertical wear < 6mm			
• 4mm < Lateral wear < 6mm	The half set of switches is safe against derailment; possible contact of wheel with switch tip could lead to increased wear.	 MEDIUM	Needs to maintenance
• 6mm < Vertical wear < 8mm			
• Lateral wear > 6mm	The half set of switches is not safe against derailment.	 BAD	Needs to change
• Vertical wear > 8mm			

10.2.3 Breakouts of Switch

Procedure: For this test the switch rail must rest on the slide chair and has to be pushed as good as possible against the stock rail with a crowbar.

It is chipped/cracked over small lengths aggregating to 200 mm. within a distance of 1000 mm. from its toe:

- (i) chipped length will be the portion where tongue rail has worn out for a depth of more than 10 mm over a continuous length of 10 mm.
- (ii) it has developed knife edged tip (thickness of top edge being less than 2 mm) over a length of more than 100 mm anywhere upto a distance of 1000 mm from its toe.

Note: Full set of half set of switch shall be changed if components needs to be replaced due to wear.

10.4 Check List for Maintenance of Turnout:

S.NO	CHECK SHEET OF SWITCH ASSEMBLY(O4756-04) FOR 1 IN 12 555R CANTED TURNOUT						
	DISCRIPTION OF ITEMS		DESIGNED VALUE mm	MEASURED VALUES ON ASSEMBLED/LAID SET			
				T.O NO.... LH/RH		T.O NO... LH/RH	
				LH	RH	LH	RH
1	Sleeper spacing in switch portion	See Separate sheet					
2	Distance between Gauge faces of outer rails	See Separate sheet					
3	Distance SRJ - ATS	T.P & SRJ at same point	745.5±6				
4	Distance ATS- 72 MM Head (JOH)	Curved/Straight (End of side Machining)	8013/8000				
5	Location of Anticreep Device	Between SL no 26-27	In Centre				
6	Straightness of Straight Stock Rail	By Thread scale &	± 01				
7	Versine of Curved Tongue rail at (4042.5, 8085, 12127.5) ± 05 (By thread & scale)	V1	44				
		V2	59				
		V3	44				
8	Straightness of Straight Tongue Rail	By Thread scale &	± 01				
9	Versine of Curved stock rail at (4225, 8450, 12675) ± 05 (By thread & scale)	V1	48				
		V2	64				
		V3	48				

11	Switch set for Main Line						
	i) Throw (If Point M/C fixed) ii) Clearance on Open side/Gap on closed side (When point M/C is fixed)	At ATS At JOH (with effective back drive)	160 60/0				
	iii) Matching of Stock & Tongue/ Rail	TOE to JOH	within 01				
12	Switch Set for Turnout						
	i) Throw (If Point M/C fixed) ii) Clearance on Open side/Gap on closed side (When point M/C is fixed)	At ATS At JOH (with effective back drive)	160 60/0				
	iii) Matching of Stock & Tongue/ Rail	TOE to JOH	within 01				
13	Vertical & Lateral Wear	Location is shown in table no: 1	OK/Not Ok				
	Condition of Tongue rails: Weather chipper or cracked	over 200mm length within 1000mm from ATS.	OK/Not Ok				
14	Track Gauge	At alternate sleepers From SL no 01 to 28	Main Line 1676 +3,-0 Turnout 1676 +3 -0				
15	Piroll with Double Roller	AT SL NO.2,5,8,11, 13,15	OK/not OK				
16	Piroll with Single Roller	AT SL NO 17,19	OK/not OK				
17	Piroll with Spring loaded Roller	AT SL NO 21,23	OK/not OK				
18	M.S Plate	At SL NO 3,16	OK/not OK				
19	Insulated Tie Plate	AT SL NO 2	OK/not OK				
20	Fittings in switch assembly	Complete fittings as per drawing	OK/not OK				

21	Fitment of Back drive	At SL NO 2,3,8,13 & 14	On opposite side of point M/C on Base plates				
22	Fitment of two point machines	Between SL NO:2-3					
23	Effectiveness of Anticreep system	Equal side gaps/tight bolts	OK/Not OK				
24	Versine of curved tongue rail (To match with curvature of lead rail) Starting from SRJ	At 6 m chord and 3 M interval	8				
25	Gap between slide block & T/rail web	At each Slide block	within 01				
26	Tightness of bolts	Visual	OK/Not OK				
27	Tightness of ERC	Visual	OK/Not OK				
28	Tightness of wedge/Leaf springs	On all slide chairs	OK/Not OK				
29	Tightness of plate screws/other fittings	On all chairs	OK/Not OK				
30	Effectiveness of 2nd point M/C		OK/Not OK				
31	Fitment of all Chairs /fittings/GRSP/CGSP on position	As per relevant drawings	OK/Not OK				
32	Deficiencies						
33	Suggestions/ Remarks						

S.NO	CHECK SHEET OF WELDABLE CROSSING ASSEMBLY FOR 1 IN 12 555R CANTED TURNOUT					
	DISCRIPTION OF ITEMS	DESIGNED VALUE mm	MEASURED VALUES ON ASEMBLED/LAID SET			
			T.O No Xing SLNO	T.O No Xing SLNO	T.O No Xing SLNO	T.O No Xing SLNO
			LH/RH	LH/RH	LH/RH	LH/RH
1	Sleeper spacing in switch portion	See Separate sheet				
2	Condition of Crossing	Ok/Not ok				
3	Wear of Crossing (Wear to be measured with straight edge at 100 mm from ANC. 1. On left wing rail 2. On Nose 3. On right wing rail	Less than 4mm (Nose) Less than 8mm (wing rail)				
4	Check rail clearance LH/RH	44				
5	Condition of Check rail	Ok/Not Ok				

11. PREVENTIVE MAINTENANCE

This part of the manual includes all of the preventive or periodical maintenance and inspection work to be done to keep the turnout in a proper condition that allows a safe operation. This part of the manual does not include the parts and actions on the signalling equipment.

The turnout is split in its main units as described in the introduction chapter – switch device, intermediate rails and crossing area with check rails with their subunits.

The part list of each turnout can also be found in the design approval from VAEVKN to check completeness and to identify part number for spare parts.

ID	Part	Component check	Criterion Limits	Action	Interval	Remark
1	Switch Device – Preventive Maintenance					
1.0.1	General Track Condition	Geometry check (Gauge, unevenness, alignment and level of the track)	Visual detectable regulations of straightness or curves and cant.	Information to track supervisor, to get information if corrective maintenance has to be done.	2 months	As per contract clause 17.9.3 Pt. 2 vol. 5 & IRPWM clause 237.
1.0.2	General Track Condition	Geometry check (Gauge, position and level of the track, horizontal and vertical)	Check with measuring tape according to Cl. 17.9.3 & IRPWM cl. 237	Information to track supervisor, to get information if corrective maintenance has to be done.	6 months	As per contract clause 17.9.3 Pt. 2 vol. 5 & IRPWM clause 237.
1.0.3	Contact surface switch blade/stock rail	Contact surface condition between stock rail and switch blade	In the drive rod axis a maximum gap of 1mm is permissible	Readjustment of point machine	2 months	
1.0.4	Contact surface switch blade/stock rail	Burr formation	Detection of burrs larger than 2mm	Grinding of affected areas	2 months	
1.0.5	Switch Stops	Visual check of connection to the stock rail, check of bolts	Existence of parts and bolts	Missing parts to be replaced, lose bolts to be tightened	2 months	Torque value- 200Nm
1.0.6	Switch Stops	Check if switch rail is in contact with switch stops	Maximum gap of blade and switch stop: 2mm	Readjustment of blade, cleaning of slide chairs	2 months	
1.0.7	Anti creep device	Visual check of position and connection to the rails, visual inspection of gap between fork and pin	Existence of parts and bolts	Missing parts to be replaced, lose bolts to be tightened	2 months	Torque value- 200Nm
1.0.8	Rail and plate fastening material	Bolts, nuts, washers etc.	Existence of parts and bolts	Missing parts to be replaced, lose bolts to be tightened	2 months	

1.0.9	Switchblades/stock rails	Clearance check at JOH	Min 60mm at JOH	Cleaning of sliding chairs, readjustment of blades	2 months	
1.0.10	Blades-contact area slide chairs	Gap between switchblade foot and slide chair	Max. 2mm	The track supervisor should be informed if failure occurs successively over a few sleepers or if a height readjustment of the sleepers has to be done or the half set of switches has to be replaced	2 months	
1.1	Stock Rails – Preventive Maintenance					
1.1.1	Running surface	Visual check for surface defects, cracks, burrs or other irregularities	Rail defects according to IRPWM Chapter 2.	Burrs to be grinded as soon as possible, other actions according to IRPWM.	2 months	
1.1.2	Rails	Defects in the rail body to be detected by ultrasonic tests (measuring car)	USFD manual	According to USFD manual	6 months	
1.1.3	Welded joints	Visual inspection of joints	Running surface defects, cracks according to FBW manual	FBW manual	2 months	
1.2	Switch rails – Preventive Maintenance					
1.2.1	Running surface	Visual check for surface defects, cracks, burrs or other irregularities	Rail defects according to IRPWM Chapter 2.	Burrs to be grinded as soon as possible	2 months	
1.2.2	Running surface	Wear of height and running surface, surface defects.	Visual check of irregularities, if necessary check with track gauges to determine derailment	As per IRPWM chapter 2.	2 months	

			safety (IRPWM Chapter 2)			
1.2.3	Running surface	Running surface and running edge (wheel contact area) in switch blade.	check with track gauges to determine derailment safety (IRPWM Chapter 2)	As per IRPWM Chapter 2	6 months	
1.2.4	Rail body	Defects in the rail body to be detected by ultrasonic tests (Measuring equipment)	USFD manual	As per USFD manual	6 months	
1.2.5	Welding joints	Visual inspection of joints	Running surface defects, cracks	As per FBW manual	2 months	
1.3	Slide Plates – Preventive Maintenance					
1.3.1	Slide chairs & rollers	Visual check	Cracks	Replacement of broken parts	2 months	
1.3.2	Leaf spring	Visual check of correct position of leaf spring		Parts to be readjusted	2 months	
1.3.3	Sliding surface & rollers	Visual check regarding excessive wear or heavy pollution		Extremely polluted chairs have to be cleaned (increased setting forces). Extremely worn plates to be replaced	2 months	
1.3.4	Mk-V clips	Visual check of correct placement, check if parts are not lost	Mk-V Clip to be assembled	Readjustment of clips, broken or missing parts to be replaced	2 months	
1.3.5	Rail pads and base plate pads	Visual check	Heavy deformation or cracks	Broken or extremely deformed parts to be replaced	2 months	

1.4 General Base plates – Preventive Maintenance						
1.4.1	Base plate	Visual check	Cracks	Replacement of broken parts	2 months	
1.4.2	Mk-V clips	Visual check of correct seat, check if parts are not lost	Mk-V Clip to be assembled	Readjustment of clips, broken or missing parts to be replaced	2 months	
1.4.3	Rail pads and base pads	Visual check	Heavy deformation or cracks	Broken or extremely deformed parts to be replaced	2 months	

ID	Part	Component check	Criterion Limits	Action	Interval	Remark
2	Intermediate Rails – Preventive Maintenance					
2.0.1	General Track Condition	Geometry check (Gauge, position and level of the track)	Visual detectable irregularities of straightness or curves respectively	Information to track supervisor, to get information if corrective maintenance has to be done.	2 months	General Track Condition
2.0.2	General Track Condition	Geometry check (Gauge, position and level of the track)	Check with measuring tape, limits according to IRPWM	Information to track supervisor, to get information if corrective maintenance has to be done.	6 months	General Track Condition
2.0.3	Running surface	Visual check for surface defects, cracks, burrs or other irregularities	Rail defects according to IRS T10 or T-12	Burrs to be grinded as soon as possible	2 months	
2.0.4	Rails	Defects in the rail body to be detected by ultrasonic tests (measuring equipment)	USFD manual	USFD manual	6 months	
2.1	General Base plates – Preventive Maintenance					
2.1.1	Base plate	Visual check	Cracks	Replacement of broken parts	2 months	
2.1.2	Mk-V clips	Visual check of correct seat, check if parts are not lost	Mk-V clip to be assembled	Readjustment of clips, broken or missing parts to be replaced	2 months	
2.1.3	Rail pads and base plate pads	Visual check	Heavy deformation or cracks	Broken or extremely deformed parts to be replaced	2 months	
ID	Part	Component check	Criterion Limits	Action	Interval	Remark

3 Crossing Area – Preventive Maintenance						
3.0.1	General Track Condition	Geometry check (Gauge, position and level of the track, horizontal and vertical)	Visual detectable irregularities of straightness or curves and cant.	Information to track supervisor, to get information if corrective maintenance has to be done.	2 months	As per contract clause 17.9.3 Part 2 vol. 5 & IRPWM clause 237
3.0.2	General Track Condition	Geometry check (Gauge, position and level of the track, horizontal and vertical)	Check with measuring according to Cl. 17.9.3 & IRPWM cl. 237	Information to track supervisor, to get information if corrective maintenance has to be done.	6 months	As per contract clause 17.9.3 Part 2 vol. 5 & IRPWM clause 237
3.0.3	Running surface	Visual check for surface defects, cracks, burrs or other irregularities	Rail defects according to IRPWM Chapter 2	Burrs to be grinded as soon as possible	2 months	
3.0.4	Rails	Defects in the rail body to be detected by ultrasonic tests (measuring equipment)	USFD manual	USFD manual	6 months	
3.1 Cast Manganese Crossing – Preventive Maintenance						
3.1.1	Running surface	Visual check for surface defects, cracks, burrs or other irregularities	Rail defects according to IRPWM Chapter 2	Burrs to be grinded as soon as possible	2 months	
3.1.2	Running surface	Wear in height and running surface, surface defects, breakouts	Visual check of irregularities, if necessary check with track gauges to determine derailment safety (see also IRPWM)	IRPWM Chapter 2	2 months	

3.1.3	Rails (Not cast manganese part of crossing)	Defects in the rail body to be detected by ultrasonic tests (measuring equipment)	USFD manual	According to internal USFD manual	6 months	
3.1.4	Welding joints	Visual inspection of joints	Running surface defects, cracks	FBW manual	2 months	
3.1.5	Welding joints – Cast manganese body	Visual inspection of joints	Cracks	Replacement of crossing	2 months	Information to VAEVKN
3.1.6	Breakouts /cracks in cast manganese body	Visual check of running surface	Breakouts / heavy deformation	Action accordingly, welding repair	2 months	
3.2	Check Rail – Preventive Maintenance					
3.2.1	Face of Check rail	Visual inspection of face of check rail	Visual check for surface defects, cracks, burrs or other irregularities	Burrs to be grinded as soon as possible	2 months	
3.2.2	Distance between the running edge and face of check rail	Inspection of distances according to VAEVKN layout	Limits shown on turnout drawing or check rail arrangement for each turnout	Readjustment with shims	2 months	
3.3	General Base plates – Preventive Maintenance					
3.3.1	Base plates	Visual check	Cracks	Replacement of broken parts	2 months	

3.3.2	Mk-V clips	Visual check of correct placement, check if parts are not lost	Mk-V clip to be assembled	Readjustment of clips, broken or missing parts to be replaced	2 months	
3.3.3	Rail pads and base plate pads	Visual check	Heavy deformation or cracks	Broken or extremely deformed parts to be replaced	2 months	
3.4	Check rail plates (fastening of check rail) – Preventive Maintenance					
3.4.1	Check rail plate	Visual check	Cracks	Replacement of broken parts	2 months	
3.4.2	Rail pads and base plate pads	Visual check	Heavy deformation or cracks	Broken or extremely deformed parts to be replaced	2 months	
3.4.3	Fastening material	Bolts, nuts, washers, etc.	Existence of parts and bolts	Missing parts to be replaced, lose bolts to be tightened	2 months	Torque value- 200Nm

12.CORRECTIVE MAINTENANCE

This part of the manual includes the corrective maintenance actions that are to be taken either during inspections (while preventive checks) or if single failures occur. The maintenance actions described in this section should bring the turnout back into a safe and adequate condition.

All parts of the turnout are listed in the tables below with their failures that could occur. Further the list shows which immediate actions or respective corrective actions are to be taken. A corrective maintenance action that could prevent a single failure with the link to the chapter preventive maintenance can also be found in the table. Especially if failures occur regularly please check if the preventive maintenance of the part is done properly and periodically.

ID	Failure	Detection	Immediate steps	Corrective maintenance	Preventive maintenance	Remark
1	Stock rails – corrective maintenance					
1.1	Cross break	Train operator (noise or impact), during visual inspection	Joining rails with fish plates, speed restriction	Replacement of half set of switches	1.1.1, 1.1.2	
1.2	Larger surface defects	Visual inspection	Increased inspection intervals	Replacement of half set of switches	1.1.1, 1.1.2	
1.3	Cracks in railhead	Visual inspection, ultrasonic tests	Action as per USFD report	Grinding of affected areas	1.1.1, 1.1.2	
1.4	Corrugation / RCF of running surface	Visual inspection	Not necessary	Grinding of affected areas	1.1.1, 1.1.2	
2	Switch stops – corrective maintenance					
2.1	Cross break	Visual inspection	In case of failure , if 1 or more breaks occur, immediate change of parts	Replacement of parts	1.0.5, 1.0.6	
2.2	Missing or broken fastening material	Visual inspection	In case of single failure or breaks occur, immediate change of parts	Replacement of parts	1.0.5, 1.0.6	

ID	Failure	Detection	Immediate steps	Corrective maintenance	Preventive maintenance	Remark
3	Anti creep device – corrective maintenance					
3.1	Cross break, missing part	Visual inspection	Plan replacement of it	Replacement of parts	1.0.7	
3.2	Missing or broken fastening material	Visual inspection	Plan replacement of it	Replacement of parts	1.0.7	
4	Switch blade – corrective maintenance					
4.1	Cross break of switch blade in moveable part of the rail	Detection by point machine	Trains must not be passed on turnout anymore	Replacement of half set of switches	1.2.1, 1.2.2, 1.2.3, 1.2.4	
4.2	Cross break in fixed area of switch rail	Train operator (noise or impact), during visual inspection	Trains must not be passed on turnout anymore	Replacement of half set of switches	1.2.1, 1.2.2, 1.2.3, 1.2.4	
4.3	Critical breakouts on running edge	Visual inspection	Check with track gauges to determine derailment safety, if necessary grinding, speed restriction & plan replacement	Replacement of half set of switches	1.2.1, 1.2.2, 1.2.3, 1.2.4	
4.4	Small breakouts on running edge	Visual inspection	Close watch	Grinding of affected areas	1.2.1, 1.2.2, 1.2.3, 1.2.4	
4.5	Large surface defects on the railhead	Visual inspection	Increased inspection intervals & close watch	Replacement of half set of switches	1.1.1, 1.1.2	
4.6	Cracks in railhead	Visual inspection, ultrasonic tests	Plan restriction and speed restriction	Grinding of affected areas	1.2.1, 1.2.2, 1.2.3, 1.2.4	
4.7	Corrugation / RCF of running surface	Visual inspection	Not necessary	Grinding of affected areas	1.2.1, 1.2.2, 1.2.3, 1.2.4	

ID	Failure	Detection	Immediate steps	Corrective maintenance	Preventive maintenance	Remark
5	Crossing – corrective maintenance					
5.1	Cross break of closure rails	Train operator (noise or impact), during visual inspection	Joining rails with fish plates, speed restriction	Replacement of crossing	3.0.3, 3.0.4, 3.0.5, 3.0.6, 3.1.1, 3.1.2, 3.1.3, 3.1.4	
5.2	Cracks or cross break in welding zone (standard rails/manganese part)	Visual inspection	Joining rails with fish plates, speed restriction	Replacement of crossing	3.0.3, 3.0.4, 3.0.5, 3.0.6, 3.1.1, 3.1.2, 3.1.3, 3.1.4	
5.3	Cross break of cast manganese part	Train operator (noise or impact), during visual inspection	Trains must not be passed on turnout anymore	Replacement of crossing		
5.4	Critical breakouts on running edge	Visual inspection	Check with track gauges to determine derailment safety, if necessary grinding, speed restriction & plan replacement	Repair welding in cast manganese part if possible, Replacement of crossing	3.0.3, 3.0.4, 3.0.5, 3.0.6, 3.1.1, 3.1.2, 3.1.3, 3.1.4	
5.5	Small breakouts on running edge	Visual inspection	Close watch	Grinding of affected areas	3.0.3, 3.0.4, 3.0.5, 3.0.6, 3.1.1, 3.1.2, 3.1.3, 3.1.4	
5.6	Large surface defects on the railhead	Visual inspection	Increased inspection intervals & close watch	Replacement of crossing	3.0.3, 3.0.4, 3.0.5, 3.0.6, 3.1.1, 3.1.2, 3.1.3, 3.1.4	

ID	Failure	Detection	Immediate steps	Corrective maintenance	Preventive maintenance	Remark
5.7	Cracks in railhead	Visual inspection, ultrasonic tests	USFD testing	Renewal as per merit of case	3.0.3, 3.0.4, 3.0.5, 3.0.6, 3.1.1, 3.1.2, 3.1.3, 3.1.4	
5.8	Corrugation / RCF of running surface	Visual inspection	Not necessary	Grinding of affected areas	3.0.3, 3.0.4, 3.0.5, 3.0.6, 3.1.1, 3.1.2, 3.1.3, 3.1.4	
6	Check rail – corrective maintenance					
6.1	Wrong check gauge dimension	Inspection or damage of crossing nose	Not necessary	Readjustment of check rail with shims	3.0.1, 3.0.2, 3.0.3, 3.0.4, 3.2.1, 3.2.2	
6.2	Missing or broken fastening material	Visual inspection	In case of failure, breaks occur, immediate change of parts	Replacement of parts	3.2.1, 3.2.2	
6.3	Excessive wear of check face	Visual inspection	Not necessary	Change of checkrail	3.2.1, 3.2.2	
7	Intermediate rails – corrective maintenance					
7.1	Cross break	Train operator (noise or impact), during visual inspection	Joining rails with fish plates, speed restriction	Replacement of rails	2.0.1, 2.0.2, 2.0.3, 2.0.4	
7.2	Larger surface defects	Visual inspection	Increased inspection intervals, close watch	Replacement of rails	2.0.1, 2.0.2, 2.0.3, 2.0.4	
7.3	Cracks in railhead	Visual inspection, ultrasonic tests	USFD testing	Renewal as per merit of case	2.0.1, 2.0.2, 2.0.3, 2.0.4	
7.4	Corrugation / RCF of running surface	Visual inspection	Not necessary	Grinding of affected areas	2.0.1, 2.0.2, 2.0.3, 2.0.4	

ID	Failure	Detection	Immediate steps	Corrective maintenance	Preventive maintenance	Remark
8	Base plates – corrective maintenance					
8.1	Cross break	Visual inspection	In case of failure or break occur, immediate change of parts	Replacement of parts	1.4.1, 1.4.4	
9	Slide plates – corrective maintenance					
9.1	Cross break	Visual inspection	In case of failure or two breaks occur, immediate change of parts	Replacement of parts	1.3.1, 1.3.2, 1.3.5	
10	Slide plates – corrective maintenance					
10.1	Heavy pollution	Visual inspection, increased setting force	When setting problems occur, immediate cleaning	Cleaning of part	1.3.3	
10.2	Excessive wear	Visual inspection, increased setting force	When setting problems occur, immediate cleaning, resp. change of base plates	Replacement of parts	1.3.3	
14	Rail pads and base plate – corrective maintenance					
14.1	Break, missing parts	Visual inspection	Not necessary	Replacement of parts	1.4.4	